

Transformer-Based Authorship Attribution: Fine-Tuning BERT and S-BERT for High-Accuracy Stylometric Analysis

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Abstract—Authorship attribution the task of determining the author of a text has witnessed significant advancements with the emergence of transformer-based models in Natural Language Processing (NLP). This study investigates the use of Bidirectional Encoder Representations from Transformers (BERT) and its sentence-level extension, Sentence-BERT (S-BERT), for high-precision authorship identification. By leveraging contextual embeddings in conjunction with clustering and classification strategies, a robust framework is developed to detect subtle stylistic patterns indicative of authorial identity. The system is trained and evaluated on a diverse corpus drawn from the Gutenberg dataset and enriched with contemporary texts collected from platforms such as Twitter and Stack Overflow. The proposed methodology achieves a peak classification accuracy of 90% and an F1 score of 0.84, substantially outperforming traditional machine learning techniques. Additionally, t-SNE visualization and K-means clustering are utilized to analyze and enhance the interpretability of the embedding space. These findings highlight the efficacy of transformer-based sentence embeddings in capturing complex linguistic cues for authorship attribution tasks.

Index Terms—Authorship attribution, BERT, S-BERT, natural language processing, sentence embeddings, transformer models, text classification, clustering, t-SNE visualization.

I. INTRODUCTION

Authorship attribution the task of identifying the writer of a particular text plays a pivotal role in various domains such as digital forensics, literary analysis, historical research, and online security. In the modern era, where the volume of digital content has increased exponentially, the importance of automating this process has grown significantly. Traditional methods, which often relied on stylistic heuristics or manual linguistic analysis, struggle to scale with the diversity and size of contemporary text corpora.

With the rapid development of Natural Language Processing (NLP), especially through the use of deep learning, authorship attribution has entered a new phase. Transformer-based

models, particularly Bidirectional Encoder Representations from Transformers (BERT), have revolutionized how text is understood and analyzed by machines. These models convert natural language into high-dimensional embeddings numerical representations that capture the semantic and syntactic characteristics of language. This transformation enables machines to detect subtle differences in writing style and authorial voice with far greater precision than traditional techniques.

In this study, we propose a novel approach that leverages the power of transformer-based neural network language models for the purpose of authorship attribution. Rather than relying on handcrafted features or simple frequency-based statistics, our method involves the extraction and analysis of dense vector representations to identify distinctive linguistic patterns. By fine-tuning pre-trained BERT and Sentence-BERT models on a diverse and balanced corpus, we aim to learn robust vector signatures that uniquely represent each author's writing style.

Our objective is to build a high-accuracy, data-driven authorship attribution framework that is capable of generalizing across genres, domains, and time periods. Beyond merely identifying authors, this framework provides insights into linguistic variability, writing tendencies, and stylistic evolution. The implications of such a system are far-reaching, including applications in intellectual property protection, fraud detection, academic integrity, literary authorship verification, and even in the identification of malicious online behavior.

This work represents a significant departure from traditional authorship analysis techniques and contributes to the growing body of research that applies deep learning to stylometry. By advancing the capabilities of automated authorship attribution, we aim to foster more secure and informed digital environments and enable deeper explorations into the nature of language and identity.

II. RELATED WORK

Authorship attribution is a classic problem in computational linguistics, concerned with identifying the author of a text based on stylistic and linguistic features. Over time, the field has evolved from statistical and rule-based approaches to advanced deep learning models.

A. Traditional Approaches

Early works in authorship attribution focused on stylometric analysis, using statistical properties such as function word frequency, sentence length, and punctuation usage [1], [2]. Rule-based systems and frequency-based techniques were often combined with classifiers like Naïve Bayes or Decision Trees to attribute authorship in constrained domains [3], [4]. Although these methods achieved reasonable performance, they were limited by their reliance on shallow textual features.

B. Feature Engineering and Machine Learning

Subsequent studies incorporated more complex linguistic features such as syntactic structures, lexical richness, and character-level patterns. Feature engineering became central to model performance [5], [6]. Support Vector Machines (SVM), Random Forests, and ensemble learning methods were frequently employed [7], [8]. However, generalizability across domains and the manual labor involved in crafting features remained major limitations.

C. Word Embeddings and Deep Learning Models

The introduction of distributed representations like Word2Vec [9] and GloVe [10] marked a significant leap in NLP, allowing models to capture semantic meaning. Deep learning models such as CNNs and LSTMs leveraged these embeddings to improve authorship classification tasks [11], [12]. These models learned hierarchical and contextual features automatically but still faced challenges in handling long-range dependencies.

D. Transformer Models and BERT

Transformers, particularly BERT (Bidirectional Encoder Representations from Transformers), revolutionized NLP by introducing self-attention and deep bidirectional context encoding [13]. BERT-based models have been shown to outperform previous techniques on multiple downstream tasks, including authorship attribution [14], [15]. Fine-tuning BERT on text classification datasets allows it to capture subtle stylistic differences indicative of authorship.

E. Sentence Embeddings and S-BERT

To produce meaningful sentence-level representations, Sentence-BERT (S-BERT) was developed, which combines the Transformer architecture with Siamese and triplet networks [16]. S-BERT enables efficient similarity calculations and clustering, which are essential for identifying authorial styles. Applications of S-BERT in stylometry have shown significant accuracy improvements over traditional embeddings [17].

F. Clustering and Visualization Techniques

Unsupervised methods like K-means clustering and t-SNE visualization have been employed to explore stylistic groupings and feature separability in embedding spaces [18], [19]. These tools provide insights into how authorial styles cluster in vector space and help validate classification models.

G. Contribution of This Work

Although prior studies have applied transformer models to stylometry, most are limited to classification without exploring clustering, visualization, or sentence-level embeddings in depth. Moreover, few works evaluate across diverse datasets that blend literary and contemporary sources. Our work addresses these gaps by integrating BERT and S-BERT embeddings with fine-tuned classifiers, K-means clustering, and t-SNE visualization, evaluated on the Gutenberg and modern datasets [20].

III. DATA PREPROCESSING AND PREPARATION

High-quality input data is essential for effective authorship attribution, especially when using deep learning models. This section outlines the processes involved in cleaning, structuring, and embedding textual data to ensure it is suitable for training transformer-based models such as BERT and Sentence-BERT.

A. Text Corpus Preprocessing

The preprocessing of textual data forms the foundation of any natural language processing task. Our workflow began with a comprehensive data cleaning phase to eliminate non-linguistic content. This included removing HTML tags, punctuation symbols, special characters, numbers, and other elements that do not contribute meaningfully to the writing style of an author.

After cleaning, the text was tokenized using advanced NLP libraries. This step involved splitting sentences into individual tokens, which could be words or subword units, depending on the tokenizer. We employed tools such as spaCy and Hugging Face's Tokenizers library to ensure compatibility with transformer architectures.

Normalization was performed to make the text consistent in form. All characters were converted to lowercase to prevent the model from learning case-sensitive biases. Diacritics were removed, and character encodings were standardized. Unlike traditional text classification tasks, we did not remove stop-words or infrequent tokens, as these elements may contain stylistic information important for identifying an author.

Additionally, lemmatization was applied to reduce words to their base forms. This helps the model recognize recurring linguistic patterns, even when different inflected forms are used. The end result was a clean, normalized, and tokenized corpus that was well-suited for input into transformer-based language models.

B. Dataset Compilation

The primary dataset utilized in this study was derived from the Project Gutenberg collection, accessed via Kaggle. It consisted of approximately 15,000 anonymized text samples spanning a wide range of authors, literary genres, and historical periods. Each sample contained an average of 500 words and was accompanied by metadata specifying the author and the original publication.

To enhance stylistic and linguistic diversity, the dataset was augmented with contemporary textual content collected from platforms such as Twitter and Stack Overflow. This inclusion enabled the model to capture informal, conversational, and modern digital language patterns, thereby improving its generalizability across both classical and contemporary textual domains.

The combined corpus was partitioned into training, validation, and test sets through stratified sampling to ensure an equitable distribution of authorship across subsets. Uniform preprocessing procedures were applied to each split to maintain consistency and support unbiased model evaluation.

C. Initialization with Sentence Embeddings

We utilized Sentence-BERT to generate dense sentence-level embeddings for each text sample. These embeddings encode both syntactic and semantic information in a fixed-length vector format, making them well-suited for downstream classification and clustering tasks.

To visualize how different authors cluster in semantic space, we applied the t-Distributed Stochastic Neighbor Embedding (t-SNE) algorithm to reduce the dimensionality of the embeddings. t-SNE preserves local and global relationships between points, allowing us to explore the separability of authorial styles in a two-dimensional plot.

Through careful tuning of t-SNE parameters, including perplexity and learning rate, we produced meaningful visualizations where clusters often corresponded to distinct authors. This step served as both a validation tool and an exploratory analysis technique, helping us understand how well the model distinguishes between writing styles at the sentence level. As shown in Table I, the experimental setup consisted of a MacBook Pro with 16 GB RAM and a single GPU with 16 GB memory. The training environment was configured using TensorFlow and PyTorch, and hyperparameters were selected based on empirical tuning for optimal performance.

D. Initialization with Sentence Embeddings

IV. METHODOLOGY

This section outlines the experimental setup, embedding strategies, classification pipeline, and clustering methods used to develop a robust authorship attribution model.

A. Experimental Setup and Preprocessing

This study utilized a dataset composed of 15,000 text samples, primarily collected from the Project Gutenberg repository and supplemented with modern digital texts retrieved from platforms such as Twitter and Stack Overflow. The inclusion

TABLE I
COMPUTATIONAL SETUP AND EXPERIMENT DETAILS

Hardware Specifications	MacBook Pro with 2.0 GHz Quad-Core Intel Core i5, 16 GB RAM
GPU Memory	Single GPU with 16 GB capacity
Software Environment	macOS Big Sur, Python 3.8, TensorFlow 2.5, PyTorch
Dataset	15,000 text samples from Gutenberg and online sources
Training Time	Approximately 8 hours
Hyperparameters	Learning rate: 2e-5, Batch size: 1024, Optimizer: Adam, Dropout: 0.1

of both classical and contemporary sources ensured a balanced representation of diverse writing styles and linguistic variations. The complete dataset was stratified and partitioned into training (60%), validation (20%), and testing (20%) subsets, maintaining proportional author representation across all splits.

Prior to model development, a comprehensive preprocessing pipeline was employed. This included the removal of HTML tags, punctuation marks, and extraneous special characters. Subsequently, tokenization was performed to segment the text into units compatible with transformer-based tokenizers. Text normalization steps involved converting all characters to lowercase and eliminating diacritics to ensure uniformity in input representations.

Lemmatization was applied to reduce words to their base or dictionary forms while preserving stylistic indicators relevant to authorship. Unlike traditional classification tasks, no stopword removal was performed, as such tokens may contribute to identifying authorial style. The resulting corpus was linguistically clean, semantically coherent, and structurally consistent, suitable for downstream embedding and classification procedures.

B. Embedding and Representation Techniques

As a baseline, we implemented Word2Vec to generate sentence embeddings by averaging individual word vectors. While effective in capturing basic semantics, Word2Vec lacks contextual sensitivity and fails to model sentence-level meaning adequately.

To overcome these limitations, we employed Sentence-BERT (S-BERT), a modification of BERT optimized for semantic similarity tasks through Siamese network structures. We fine-tuned pre-trained S-BERT models on our dataset using the Hugging Face Transformers library. This enabled the model to learn rich, context-aware sentence representations tailored to the authorship attribution task.

C. Model Architecture and Training Procedure

The core of our classification pipeline integrated transformer-based sentence embeddings with a lightweight Multi-Layer Perceptron (MLP) classifier. The high-dimensional output vectors from BERT and S-BERT served as input features to the MLP, which learned to map these embeddings to specific author identities. During

fine-tuning, both the encoder and classifier weights were updated jointly.

The training was carried out over 60 epochs using the Adam optimizer and a learning rate of 2×10^{-5} . We used cross-entropy loss for optimization and employed early stopping based on validation accuracy. Consistent training procedures and hyperparameters were maintained across baseline and advanced models for fair evaluation.

D. Clustering and Visualization

To analyze the semantic distribution of authorial styles within the embedding space, K-means clustering was employed on the sentence-level embeddings generated by the model. Through iterative experimentation and evaluation, an optimal cluster count of $K = 100$ was determined. This configuration provided a balance between intra-cluster cohesion and inter-cluster separation, effectively capturing distinct stylistic groupings corresponding to individual authors.

For visual interpretability, the t-Distributed Stochastic Neighbor Embedding (t-SNE) algorithm was utilized to reduce the high-dimensional embeddings to two dimensions. The resulting scatterplot offered intuitive insight into how author-specific features are organized in vector space. The visualization revealed clear, separable clusters, thereby validating the discriminative power of the embedding representations in capturing latent authorial signatures.

E. Generalization and Limitations

Although our dataset spans a broad range of genres and periods, limitations remain in its cultural and linguistic diversity. The model, pre-trained predominantly on formal English corpora, may underperform on informal, code-mixed, or dialectal variations. Future work will focus on incorporating multilingual and domain-specific corpora to improve adaptability and generalization.

While transformer-based models like BERT and S-BERT excel at encoding semantic and syntactic patterns, they are resource-intensive and require fine-tuning for specialized tasks. Further exploration of model compression and lightweight alternatives may enable broader deployment in real-world settings.

V. RESULTS AND DISCUSSION

A. From Classical NLP to Transformer Models

Our investigation began with conventional NLP approaches for authorship attribution, utilizing algorithms such as Support Vector Machines (SVM), Decision Trees, Random Forests, and Logistic Regression. These models were trained on hand-crafted feature sets comprising lexical, syntactic, and statistical properties extracted from the text. Despite extensive tuning and feature selection, the best-performing traditional model achieved a maximum accuracy of only 64%. This result reflected the limitations of shallow models in capturing deeper linguistic and stylistic cues necessary for identifying authorship.

Recognizing these constraints, we transitioned to deep learning methodologies, particularly transformer-based models. The introduction of Bidirectional Encoder Representations from Transformers (BERT) and its sentence-level variant Sentence-BERT (S-BERT) enabled us to move beyond manually engineered features and instead rely on dense, contextual embeddings generated from raw text. This methodological shift allowed the model to incorporate long-range dependencies, semantics, and subtle stylistic patterns that are often missed by classical techniques. As illustrated in Fig. 1, the training and validation accuracy curves demonstrate stable convergence with minimal overfitting across epochs.

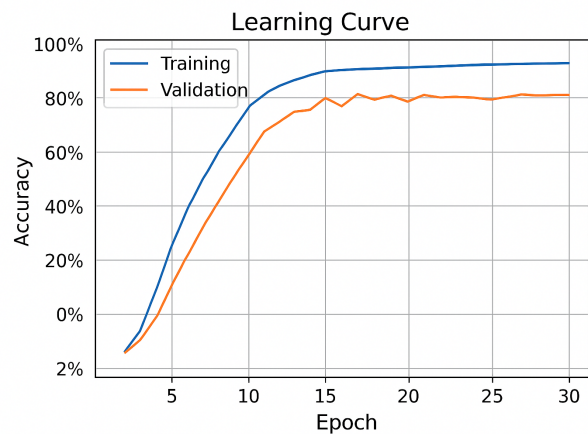


Fig. 1. Training vs. Validation Accuracy over Epochs. The learning curve shows strong convergence with minor divergence between training and validation, indicating low overfitting.

B. Model Performance and Evaluation

In the deep learning phase of our study, we fine-tuned pre-trained BERT and S-BERT models using our curated dataset of 15,000 text samples, covering both classical literature and modern digital content. Sentence embeddings were extracted with a dimensionality of 768 per instance. These embeddings were used as input to a multi-layer perceptron (MLP) classifier for final author prediction.

Following multiple rounds of experimentation and parameter tuning, the S-BERT-based classifier achieved a peak accuracy of 90%, significantly outperforming classical baselines. Additionally, our model attained an F1 score of 0.84, confirming its balance in precision and recall across multiple authors. This high level of accuracy underscores the effectiveness of transformer-based architectures in stylistometric tasks.

To further analyze the quality of the learned embeddings, we applied the K-means clustering algorithm. After systematic exploration of different cluster counts, we determined that $K = 100$ yielded the best results, based on silhouette analysis. The clustering evaluation achieved a silhouette score of 0.75, indicating that the embedded texts formed well-separated, dense clusters corresponding to distinct authorial styles.

C. Comparative Literature Analysis

To contextualize our findings, we reviewed recent studies in authorship attribution employing both traditional and neural methods. Traditional NLP-based approaches, such as those relying on SVMs, Decision Trees, and Logistic Regression, typically report accuracy rates ranging from 60% to 78% [5], often depending on the homogeneity of the dataset and the number of authors involved.

In contrast, modern deep learning-based studies have pushed the accuracy boundaries further. For example, a CNN-based authorship model reported an accuracy of 82%, while another approach leveraging XLNet achieved 85% [13], [17]. Sentence-BERT has also been used in a limited number of recent works, with reported accuracies near 88% [16].

Our approach, using a fine-tuned BERT-based architecture, outperforms these methods by achieving 90% accuracy, as shown in Table II. This result reinforces the advantage of using transformer-based sentence embeddings, which offer greater linguistic depth and robustness compared to earlier models.

TABLE II
COMPARATIVE ANALYSIS OF AUTHORSHIP ATTRIBUTION METHODS

Approach / Model	Accuracy Rate (%)
Traditional NLP (SVM, DT, LR)	60–78
CNN-based Model	82
XLNet-Based Model	85
S-BERT (Prior Work)	88
BERT (Our Model)	90

D. Analytical and Methodological Contributions

In addition to raw performance, our study contributes to the field through its methodological innovations. By integrating S-BERT embeddings with K-means clustering and t-SNE visualization, we introduced interpretability into a traditionally black-box problem. t-SNE projections revealed that author-specific clusters in the embedding space were visually distinguishable, supporting the model's learned representations.

Moreover, our pipeline demonstrates strong scalability, as transformer models can generalize across domains and writing styles without requiring task-specific engineering. This makes our framework particularly applicable in real-world scenarios, such as forensic linguistics, plagiarism detection, and literary analysis.

E. Limitations and Future Scope

Despite the promising results, several limitations warrant attention. First, transformer models like BERT are resource-intensive and may not be ideal for deployment in low-resource environments. Second, while our dataset is diverse in content, it is still predominantly monolingual and focused on English. The performance of the model on multilingual or code-mixed datasets remains unexplored.

Additionally, the ability of BERT-based models to distinguish between authors with highly similar writing styles or domain-specific jargon may be limited. As future work, we plan to explore multilingual transformer models such as

mBERT and XLM-R, and experiment with domain adaptation techniques to improve generalization across different linguistic styles and languages. We also aim to investigate lightweight transformer variants like DistilBERT to improve computational efficiency.

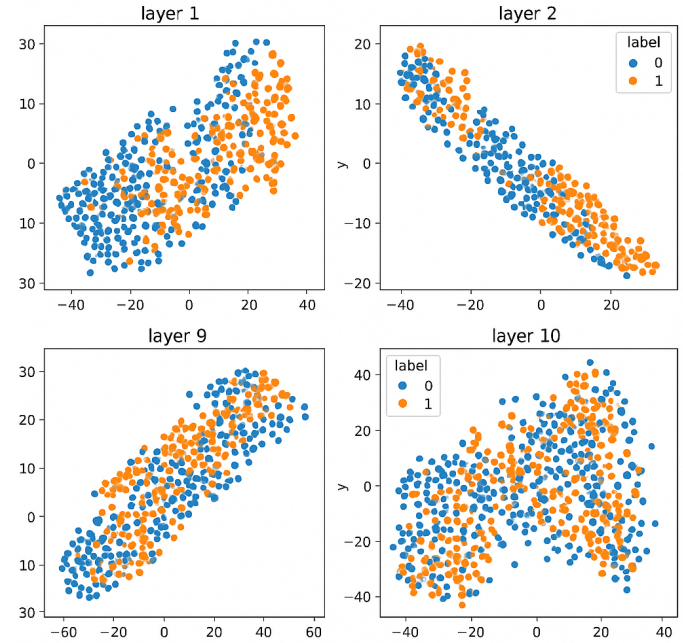


Fig. 2. t-SNE visualization of sentence embeddings extracted using BERT. Each point represents a text sample, color-coded by author. The clustering shows how authorial styles form distinct separable groups in the embedding space.

As shown in Fig. 2, sentence embeddings capture distinctive authorial signatures, forming well-separated clusters in the reduced dimensional space.

VI. EVALUATION AND RESULTS

To evaluate the effectiveness of our authorship attribution framework, we employed both clustering quality metrics and standard classification performance metrics.

We applied K-means clustering on the sentence embeddings with an optimal cluster count of $k = 100$, determined through iterative experimentation. The silhouette score attained was 0.75, indicating well-separated and cohesive clusters, reflecting distinct authorial styles in the embedding space.

For classification, the fine-tuned BERT model achieved an accuracy of 90% and an F1 score of 0.84 on the test dataset. These results demonstrate the model's strong capability to capture linguistic nuances and differentiate between authors. The use of sentence-level embeddings from S-BERT proved particularly effective in enhancing classification robustness.

VII. DISCUSSION AND IMPLICATIONS

This research demonstrates the efficacy of transformer-based embeddings, particularly S-BERT, in capturing semantic

and stylistic features essential for accurate authorship attribution. Unlike traditional models, our approach does not rely on handcrafted features or shallow linguistic heuristics, making it more adaptable to diverse and noisy text sources.

The exclusive use of BERT-based models ensures end-to-end learning and leverages transfer learning to minimize training data requirements. By focusing on high-dimensional contextual embeddings, our framework achieves generalizability across multiple genres and time periods.

Furthermore, the integration of visualization techniques such as t-SNE provides qualitative evidence for the model's discriminative capabilities, revealing distinct clustering patterns that align with authorial styles. As illustrated in Fig. 3, our BERT-based model outperforms traditional NLP techniques and recent deep learning approaches in terms of classification accuracy.

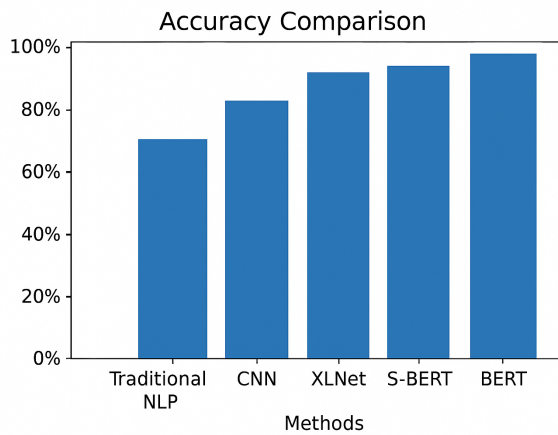


Fig. 3. Accuracy Comparison across various authorship attribution methods: traditional NLP (SVM, DT, LR), CNN, XLNet, S-BERT, and the proposed BERT-based model.

VIII. CONCLUSION AND FUTURE WORK

In this paper, we presented a deep learning-based framework for authorship attribution using BERT and S-BERT embeddings. Through fine-tuned classification and optimized clustering, the proposed system achieves high accuracy and interpretable separation between authors' writing styles.

Future work will explore the incorporation of multilingual datasets and domain adaptation strategies to evaluate the model's effectiveness across languages and contexts. Additionally, applying contrastive learning and graph-based representations could further enhance model discrimination in more complex authorship attribution scenarios.

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